TERM I EXAMINATION 2025-26
CLASS : XI
SUBJECT : MATHEMATICS

TIME: 3 HRS. **M.M.: 80**

GENERAL INSTRUCTIONS:

1- This question paper contains five sections A,B,C,D and E. Each section is compulsory. However, there are internal choices in some questions.
2- Section A has 18 MCQs and 2 Assertion - Reason based questions of 1mark each.
3- Section B has 5 very short answer type questions of 2 marks each.
4- Section C has 6 short answer type questions of 3 marks each.
5- Section D has 4 long answer type questions of 5mark each.
6- Section E has 3 source based / case based / passage based / integrated units of assessment (4 marks each) with sub parts.

SECTION - A

(Multiple Choice Questions) Each question carries 1 mark.

1. Which of the following is not correct?
(a) $\cos\theta = 1$ (b) $\sec\theta = 1/2$ (c) $\tan\theta = 20$ (d) $\sin\theta = -1/5$

2. For any integer k, the values of i^{4k+2}, i^{4k+3} is
(a) 1,-1 (b) -1,1 (c) i,-i (d) -1,-i .

3. Let R be a relation from a set A to a set B, then
(a) $R = A \cup B$ (b) $R = A \cap B$ (c) $R \subseteq A \times B$ (d) $R \subseteq B \times A$

4. If $A \cap B = \varnothing$, then
(a) $A \subseteq B$ (b) $B \subseteq A$ (c) A=B (d) None of these.

5. The radius of the circle whose arc of length 15π cm makes an angle of $3\pi/4$ radian at the centre is
(a) 10 cm (b) 20 cm (c) $11\frac{1}{4}$ cm (d) $22\frac{1}{2}$ cm

6. If $z = (3+\sqrt{2i})$ then $z\bar{z} = ?$
(a) 5 (b) 7 (c) 11 (d) $\sqrt{11}$

7. If x is a real number and $|x| < 5$, then
(a) $x \geq 5$ (b) $-5 < x < 5$ (c) $x \leq -5$ (d) $-5 \leq x \leq 5$

8. The number of ways in which 6 men can be arranged in a row so that three particular men are consecutive, is
(a) 4! X 3! (b) 4! (c) 3! X 3! (d) none of these

9. Let $A = \{x : x \in R, x > 6\}$ and $B = \{x : x \in R, x < 9\}$. Then, $A \cap B$ is equal to :
(a) (6,9] (b) (6,9) (c) [6,9) (d) [6,9]

10. The number of the proper subset of {a,b,c} is:
(a) 3 (b) 8 (c) 6 (d) 7

11. $\left(\frac{1+2i}{1-i}\right)$ lies in
(a) quadrant I (b) quadrant II (c) quadrant III (d) quadrant IV

12. If a, b and c are in G.P. then the value of $\frac{a-b}{b-c}$ is equal to.....

(a) r (b) 1/r (c) 1 (d) none of these

13. Let n(A) = m and n(B) = n. Then the total number of non-empty relations that can be defined from A to B is

(a) m^n (b) n^m-1 (c) m^n-1 (d) $2m^n$-1

14. The value of tan 1° tan 2° tan 3°.....tan 89° is

(a) 0 (b) 1 (c) 1/2 (d) not defined

15. If $\left(\frac{1+i}{1-i}\right)^n$, then n =

(a) 2m+1 (b) 4m (c) 2m (d) 4m+1, Where $m \in N$

16. If -3x + 17 < -13, then

(a) $x \in (10, \infty)$ (b) $x \in [10, \infty)$ (c) $x \in (-\infty, 10]$ (d) $x \in [-10, 10)$

17. If R is a relation on a finite set having n elements, then the number of relations on A is

(a) 2^n (b) 2^n -2 (c) 2^{n^2} (d) n^n

18. The greatest value of sin$x$.cos$x$ is

(a) 1 (b) 1/2 (c) 0 (d) -1

ASSERTION - REASON BASED QUESTIONS:

In the following questions, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices.

a. Both A and R are true and R is the correct explanation of A.

b. Both A and R are true but R is not the correct explanation of A.

c. A is true but R is false.

d. A is false but R is true.

19. **ASSERTION (A):** For $z = 3 - i\sqrt{2}, |z| = \sqrt{11}$

REASON (R): $|z| = \sqrt{x^2 + y^2}$, *if* $z = x + iy$

20. **ASSERTION (A) :** $\sin x = -\frac{1}{3}$, *then* $\cos x = \frac{2\sqrt{2}}{3}$

REASON (R): If the value of cos x is positive and sin x is negative, then $x \in \left(\frac{3\pi}{2}, 2\pi\right)$

SECTION - B

This section comprises of very short answer type- questions (VSA) of 2 marks each.

21. If the first and nth terms of a G.P. are a and b respectively and if P is the product of the first n terms, prove that $P^2 = (ab)^n$.

22. Find the domain and range of the function f(x) given by f(x) = $\frac{x-2}{3-x}$

23. *if* $\tan x = \frac{3}{4}, \pi < x < \frac{3\pi}{2}$, *find the value of* $\sin\frac{x}{2}, \cos\frac{x}{2}$

OR

Prove that : $\cos 4x = 1 - 8\ \sin^2 x \cos^2 x$

24. Find the multiplicative inverse of 2-3i.

OR

Express $(1-2i)^{-3}$ in the standard form a + ib.

25. The English alphabet has 5 vowels and 21 consonants. How many words with two different vowels and 2 different consonants can be formed from the alphabet?

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. For complex values of z, solve $|z| + z = (2 + i)$.

27. Evaluate $(\sqrt{3}+\sqrt{2})^6 - (\sqrt{3}-\sqrt{2})^6$

28. How many numbers greater than 1000000 can be formed by using the digits 1,2,0,2,4,2,4?

OR

In how many ways can the letters of the word ASSASSINATION be arranged so that all the S's are together?

29. A solution is to be kept between 40° C to 45°C . What is the range of temperature in degree Fahrenheit, if the conversion formula is $F = \frac{9}{5}C + 32$?

OR

Find all the pairs of consecutive, even positive integers, both of which are larger than 5 such that their sum is less than 23 .

30. Prove that : $\sin 3x + \sin 2x - \sin x = 4 \sin x \cos\frac{x}{2} \cos 3\frac{x}{2}$

OR

Prove that : $\cos\left(\frac{3\pi}{4}+x\right) - \cos\left(\frac{3\pi}{4}-x\right) = -\sqrt{2}\sin\sin x$

31. Using properties of sets show that $(A \cap B')' \cup (B \cap C) = A' \cup B$

Venn Diagram

SECTION - D

This section comprises of long answer - type questions (LA) of 5 marks each.

32. Expand using Binomial Theorem $(1 + x + x^2)^3$.

33. Prove that: $\cos 5A = 16\cos^5 A - 20\cos^3 A + 5\cos A$.

OR

Prove that: $\frac{1+\cancel{\sin}\sin A - \cancel{\cos}\cos A}{1+\cancel{\sin}\sin A + \cancel{\cos}\cos A} = \tan\frac{A}{2}$

34. A manufacturer has 600 litres of a 12% solution of acid. How many litres of a 30% acid solution must be added to it so that acid content in the resulting mixture will be more than 15% but less than 18%?

35. The sum of two numbers is six times their geometric mean, show that the numbers are in the ratio $(3+2\sqrt{2}) : (3-2\sqrt{2})$.

OR

If Sn denotes the sum to n terms of a G.P., prove that $(S_{10} - S_{20})^2 = S_{10}(S_{30} - S_{20})$.

SECTION - E

This section comprises of 3 case - study / passage - based questions of 4 marks each with sub- parts. First two case study questions have three sub- parts (i), (ii), (iii) of marks 1, 1, 2 respectively. The third case study question has two sub - parts of 2 marks each.

Case - Study 1 : Read the following passage and answer the questions given below.

A state cricket authority has to choose a team of 11 members, to do it so the authority asks 2 coaches of a government academy to select the team members that have experience as well as the best performers in last 15 matches. They can make up a team of 11 cricketers amongst 15 possible candidates.

Based on this information answer the following questions:

1- In how many ways can the final eleven be selected from 15 cricket players if there is no restriction?

2- In how many ways can the final eleven be selected from 15 cricket players if one of them, who is in bad form, must always be excluded?

3- In how many ways can the final eleven be selected from 15 cricket players if two of them being leg spinners, one and only one leg spinner must be included?

OR

If there are 6 bowlers, 3 wicket-keepers and 11 batsmen in all. Find the number of ways in which a team of 4 bowlers, 2 wicket-keepers and 5 batsmen can be chosen.

Case - Study 2 : Read the following passage and answer the questions given below.

In Rangoli competition in school, Preeti made Rangoli in the equilateral shape. Each side of an equilateral triangle is 24cm. The mid-point of its are joined to form another triangle . This process is going continuously infinite.

Based on above information, answer the following questions.

1- Find the side of the fifth triangle (in cm).

2- Find the area of third triangle (in sq. cm.).

3- Find the sum of perimeter of first 6 triangles(in cm.).

OR

Find the sum of area of first 6 triangles(in sq. cm.).

Case - Study 3: Read the following passage and answer the questions given below.

Seemais doing a designing course. She is working on Cartesian Art now a days. For it she wants to take x-coordinate from the set A = { 0,1,2,3,5} and y- coordinate from the set B = { -3,-2,-1,0,1,2,3} .

Based on this information answer the following questions:

i) (a) How many ordered pairs Seema can make from A to B ?

(b) How many total relations can be defined from the set A to the set B ?

ii) If a relation from A to B is defined as

$R = \{(a, b): a< b, a \in A, b \in B \}$.Then

a) Write R in roster form. Draw its arrow diagram.

b) Write domain, and range of R .